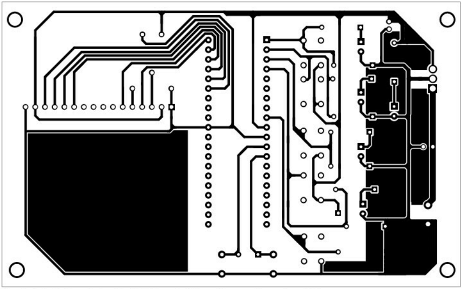

Fig. 3: Actual-size, single-side PCB layout of the voting machine using AVR

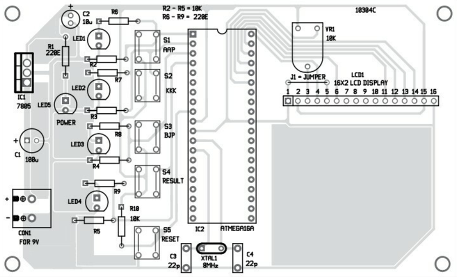

Fig. 4: Components layout for the PCB

PARTS LIST

Semiconductors:
- IC1 - 7805, 5V voltage regulator
- IC2 - ATmega16A microcontroller
- LED1-LED5 - 5mm LED

Resistors (all 1/4-watt, ±5% carbon):
- R1, R6-R9 - 220-ohm
- R2-R5, R10 - 10-kilo-ohm
- VR1 - 10-kilo-ohm preset

Capacitors:
- C1 - 100µF, 25V electrolytic
- C2 - 10µF, 16V electrolytic
- C3, C4 - 22pF ceramic disc

Miscellaneous:
- CON1 - 2-pin terminal connector
- S1-S5 - Tactile switch
- LCD1 - 16×2 LCD
- XTAL1 - 8MHz crystal oscillator
- - 40-pin IC base for ATmega16A
- - 9V battery

mode. Its data pins (D4, D5, D6 and D7) are connected to Port A (PA2, PA3, PA4 and PA5) of IC2. Control pins RS and EN are connected to pins PA0 and PA1 of IC2, respectively.

The LCD displays each character in a 5 × 7 pixel matrix. You can adjust the contrast of the LCD using a 10-kilo-ohm preset. While programming ATmega16A, note the fuse bit settings. Low fuse bit value is E4 and high fuse bit value is D9.

Construction and testing

An actual-size PCB layout for the voting machine using AVR is shown in Fig. 3 and its components layout in Fig. 4.

Connect LCD1 to the PCB as shown in the figure. Now, mount all the switches (S1 through S5) and LED1 through LED5 on the front panel of this cabinet.

Software

The software is written in 'C' language and compiled using Keil software. You can use any suitable software for programming the ATmega16A microcontroller. ProgISP programmer was used for programming at EFY Lab.

Assembly and testing

After assembling the circuit on the PCB, check it for proper connections. Now, burn the program (voting code.hex) into the microcontroller using the programmer. Insert the microcontroller into the IC base and connect 9V battery. LCD1 will show "press any key" message.

This circuit has provision to cast votes for three candidates as it uses switches corresponding to three candidates only (S1 for AAP, S2 for KKK and S3 for BJP). Switch S4 is pressed to calculate the results.

When you press any of switches S1 through S3, the corresponding pin of the microcontroller gets pulled up to Vcc and it operates as per the programmed logic. The LCD1 will show a "Thank you" message. The respective LED (LED1-LED3) must glow to indicate that voting is successful.

Press switch S4 to get results like the winner and its share of the votes. In case two or more parties get equal votes, press switch S4 again to get further details. EFY

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*

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Kit of this project is available at www.kitsNspares.com